

MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 6.4 Notes

Transition Matrices and Similarity

Devotional Thought

Linear Transformations and Change of Basis

In Section 6.3, we saw that the matrix for a linear transformation $T : V \rightarrow V$ depends on the basis for V . In other words, the matrix for T relative to a basis B is different from the matrix for T relative to another basis B' .

Our goal is to see how the matrices for a linear transformation relative to two different bases are related. Recall that in Section 4.7 we learned how to find a transition matrix which converts, via matrix multiplication, a coordinate matrix in one basis into the corresponding coordinate matrix in the other basis.

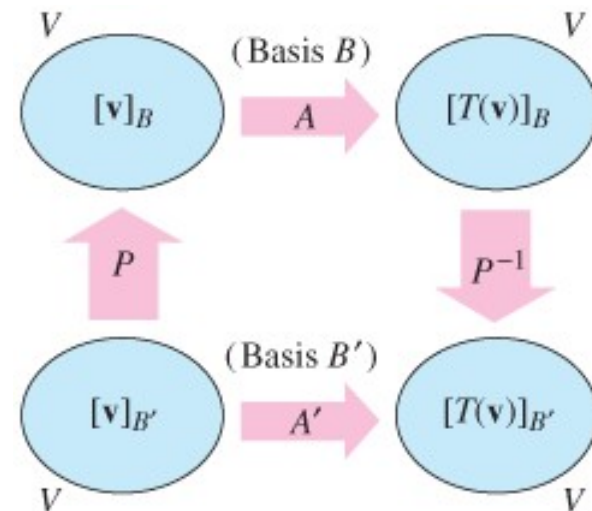
In this section, A , A' , P , and P^{-1} represent the four square matrices listed below.

Matrix for T relative to B : A

Matrix for T relative to B' : A'

Transition matrix from B' to B : P

Transition matrix from B to B' : P^{-1}



Note in the figure above there are two ways to get from the coordinate matrix $[\mathbf{v}]_B$ to the coordinate matrix $[\mathbf{v}]_{B'}$. One way is direct, using the matrix $[T(\mathbf{v})]_{B'}$ to obtain

$$A' [\mathbf{v}]_B = [T(\mathbf{v})]_{B'}.$$

The other is indirect, using the matrices P , A , and P^{-1} to obtain

$$P^{-1}AP [\mathbf{v}]_{B'} = [T(\mathbf{v})]_{B'}.$$

Example [Exercise 14]

Consider the bases $B = \{(1, 1), (-2, 3)\}$ and $B' = \{(1, -1), (0, 1)\}$ for R^2 , and let

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 4 \end{bmatrix}$$

be the matrix for $T : R^2 \rightarrow R^2$ relative to B .

- Find the transition matrix P from B' to B .
- Use the matrices P and A to find $[\mathbf{v}]_B$ and $[T(\mathbf{v})]_B$, where $[\mathbf{v}]_{B'} = \begin{bmatrix} 1 & -3 \end{bmatrix}^T$.
- Find P^{-1} and A' (the matrix for T relative to B').
- Find $[T(\mathbf{v})]_{B'}$ two ways.

Example [Exercise 18]

Consider the bases $B = \{(1, 1, -1), (1, -1, 1), (-1, 1, 1)\}$ and $B' = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for R^3 , and let

$$A = \begin{bmatrix} \frac{3}{2} & -1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{5}{2} \end{bmatrix}$$

be the matrix for $T : R^3 \rightarrow R^3$ relative to B .

- Find the transition matrix P from B' to B .
- Use the matrices P and A to find $[\mathbf{v}]_B$ and $[T(\mathbf{v})]_B$, where $[\mathbf{v}]_{B'} = \begin{bmatrix} 2 & 1 & 1 \end{bmatrix}^T$.
- Find P^{-1} and A' (the matrix for T relative to B').
- Find $[T(\mathbf{v})]_{B'}$ two ways.

Similar Matrices

Definition For square matrices A and A' of order n , A' is similar to A when there exists an invertible matrix P such that $A' = P^{-1}AP$.

Examples In our last two examples, looking at the matrix of T relative to B' we can see that A' is similar to A :

- From Exercise 14,

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 4 \end{bmatrix} \quad \text{and} \quad A' = \begin{bmatrix} 3 & 0 \\ -2 & 4 \end{bmatrix}$$

are similar because $A' = P^{-1}AP$, where $P = \begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ -\frac{2}{5} & \frac{1}{5} \end{bmatrix}$.

- From Exercise 18,

$$A = \begin{bmatrix} \frac{3}{2} & -1 & -\frac{1}{2} \\ -\frac{1}{2} & 2 & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{5}{2} \end{bmatrix} \quad \text{and} \quad A' = \begin{bmatrix} \frac{1}{4} & -1 & -\frac{5}{4} \\ -\frac{1}{4} & 2 & -\frac{1}{4} \\ \frac{5}{4} & 1 & \frac{15}{4} \end{bmatrix}$$

are similar because $A' = P^{-1}AP$, where $P = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$.

Example [Exercise 22] Use the matrix P to show that the matrices A and A' are similar.

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad A' = \begin{bmatrix} 5 & 2 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Remark Notice that A is diagonal; more on this later...

Sidenote: Similarity is an Equivalence Relation

Theorem 6.13 Properties of Similar Matrices

Let A , B , and C be square matrices of order n . Then the properties below are true.

1. A is similar to A .
2. If A is similar to B , then B is similar to A .
3. If A is similar to B and B is similar to C , then A is similar to C .

Remarks These three properties form a mathematical definition for what equality means:

- Property 1 says that similarity is **reflexive** on the set of square matrices of order n . Namely, every square matrix is similar to some matrix, namely itself.
- Property 2 says that similarity is **symmetric**: If A' is similar to A , then it is also true that A is similar to A' . So, it makes sense to simply say that **A and A' are similar**.
- Property 3 says that similarity is also **transitive**.

Put together, these three properties say that similarity is an **equivalence relation** on the set of square matrices of order n .

Proof of Theorem 6.13. Let A , B , and C be square matrices of order n .

Consider $P = I_n$. Then $P^{-1} = I_n$, meaning that

$$A = I_n A I_n = P^{-1} A P.$$

So property 1 holds.

Now assume that A is similar to B . Then there exists an invertible matrix P such that $A = P^{-1} B P$. Then

$$B = I_n B I_n = (P P^{-1}) B (P P^{-1}) = P (P^{-1} B P) P^{-1} = Q^{-1} A Q,$$

where $Q = P^{-1}$ is invertible. That is, B is similar to A . So property 2 holds.

Finally, suppose that A is similar to B and B is similar to C . Then there exist invertible matrices R and S such that $A = R^{-1} B R$ and $B = S^{-1} C S$. Then

$$A = R^{-1} B R = R^{-1} (S^{-1} C S) R = (R^{-1} S^{-1}) C (S R) = U^{-1} C U,$$

where $U = S R$ is invertible. That is, A is similar to C . So property 3 holds. ■

Two Additional Properties of Similar Matrices

[Exercise 25] Prove that if A and B are similar matrices, then $|A| = |B|$.

Example From an earlier example:

Two Additional Properties of Similar Matrices

[Exercise 25] Prove that if A and B are similar matrices, then there exists a matrix P such that $B^k = P^{-1}A^kP$ for any positive integer k .

We will see an example of why this is helpful momentarily.

An Introduction to Diagonalization

We have seen that the matrix for a linear transformation $T : V \rightarrow V$ depends on the basis used for V . This observation leads naturally to the question: What choice of basis will make the matrix for T as simple as possible? Is it always the *standard* basis? Not necessarily, as the next example demonstrates.

Example [Example 5] Let $A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$ be the matrix for $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ relative to the standard basis. Find the matrix for T relative to the basis $B' = \{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}$

Diagonal matrices have many computational advantages over nondiagonal matrices.

Also, a diagonal matrix is its own transpose. Moreover, if all of the main diagonal entries of a diagonal matrix are nonzero, then the inverse of the matrix is also a diagonal matrix, whose main diagonal entries are the reciprocals of corresponding entries in the original matrix. With such computational advantages, it is important to find ways (if possible) to choose a basis for V such that the transformation matrix is diagonal. You will pursue this problem in the next chapter.